

CoRoT-2b

R-2060

Benchmark run · workflow W8-benchmark-deep · record deep-bench_CoRoT-2_110511 · created 2026-08-09T01:01:27+00:00

BENCHMARK: NON-DETECTION

Results

Mid-Transit Time (Tmid)	2455692.9378 +/- 0.0011 BJD_TDB
Ratio of Planet to Stellar Radius (Rp/R*)	0.176 +/- 0.063
Transit depth (Rp/Rs)^2	3.11 +/- 2.23 [%]
Orbital Inclination (inc)	84.53 +/- 2.08
Airmass coefficient 1 (a1)	1.51 +/- 0.21
Airmass coefficient 2 (a2)	-0.33 +/- 0.21
Scatter in the residuals of the lightcurve fit is	13.72 %
Best Comparison Star	#2 - [510, 91]
Optimal Aperture	4.02
Optimal Annulus	9.72
Transit Duration (day)	0.079 +/- 0.018

Accuracy vs published ephemeris (ExoClock/NEA)

Rp/R■ fit vs published	0.176 ± 0.063 vs 0.1667 (deviation 0.15σ)
Duration fit vs published	0.079 d vs 0.095 d (deviation 0.89σ)
Mid-time O-C	-1.7 min
Depth SNR	2.8
Residual scatter	13.72 %

Light curve

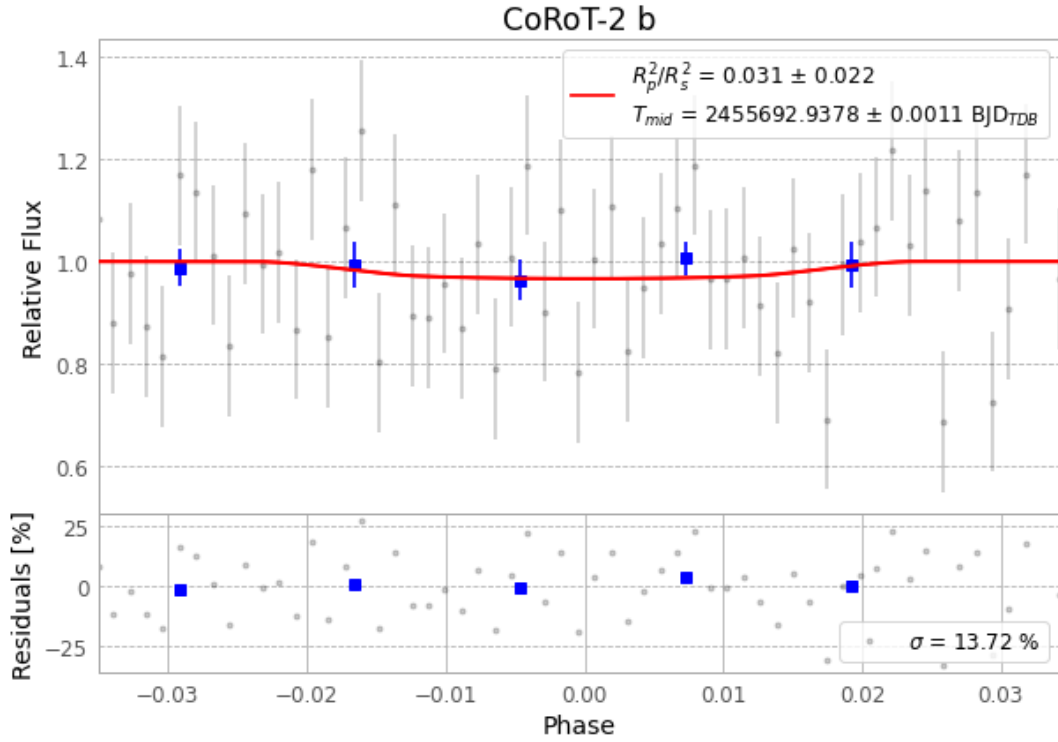


Figure 1 — FinalLightCurve_CoRoT-2 b_2011-05-11.png (SHA-256 cf44895d57844287...)

Method & software

EXOTIC 4.3.1 aperture photometry with nested-sampling transit fit (ultranest 3.6.5); priors from the NASA Exoplanet Archive. Camera CCD, binning 1x1, filter CV, target pixel [302, 178], 3 comparison star(s). Exact configuration: parameters.inits_json in record.json (SHA-256 d8834e91218f0b8b...).

exotic 4.3.1 · astropy 6.1.7 · photutils 2.0.2 · numpy 1.26.4 · scipy 1.14.1 · twirl 0.5.1 · astroquery 0.4.11 · ultranest 3.6.5 · python 3.12.13 · pipeline_commit fd74a91

Data provenance

62 FITS frames (41 MB), CoRoT-2110511085744.FITS → CoRoT-2110511120013.FITS. Every input frame is SHA-256-fingerprinted in record.json; raw frames available on request and verifiable against that manifest. Artifacts: bench-CoRoT-2-110511.log (1fc626372f96...), FinalLightCurve_CoRoT-2 b_2011-05-11.png (cf44895d5784...), FinalLightCurve_CoRoT-2 b_2011-05-11.csv (a70b42ce43f1...)

Compute resources

Wall time 201.4 s · peak memory None MB · host mac-mini-m1-8gb (macOS-26.4-arm64-arm-64bit)

Observer: — · AAVSO obscode ABGA · Automated pipeline with human review — nothing is submitted without a human approving this specific result. Data license CC-BY-4.0. Generated 2026-08-09 01:01Z.